Lab 06 – Two-sample *t*-test

ENVX1002 Handbook

Semester 1, 2026

Learning outcomes

- Learn to use R to calculate a 2-sample *t*-test
  - independent samples with constant variance
  - independent samples with unequal variance
  - paired samples
  - data transformations
- Apply the steps for hypothesis testing from lectures
- Learn how to interpret statistical output

Before you begin

=== Turn off GitHub Copilot **It is quite important that you go through this lab without using GitHub Copilot.**

It is a great tool, but it can lead to dependency on AI assistance rather than developing your statistical reasoning skills. Turn it on again after the lab if you want to use it for other projects. GitHub Copilot is also not allowed in the mid-semester exam (where you will be required to apply statistics in R, under exam conditions). To turn off GitHub Copilot:

1. Open the Command Palette (Ctrl+Shift+P or Cmd+Shift+P on Mac)
2. Type "Copilot" and select "GitHub Copilot: Enable Copilot" to disable it (make sure the checkbox is unchecked).

You can also disable it in the Settings by going to Tools > Global Options > GitHub Copilot and unchecking the "Enable GitHub Copilot" option.

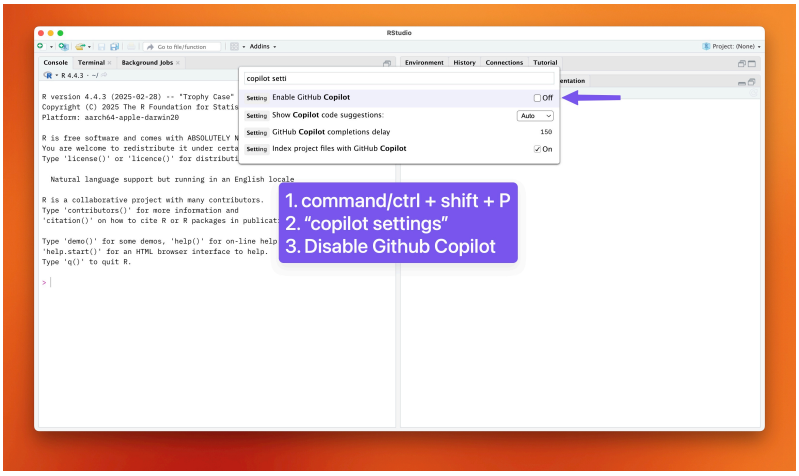


Figure 1: Disable GitHub Copilot. Click to enlarge.

Your Quarto document

Create your Quarto document and save it as `Lab-06.Qmd` or similar. The following data files are required:

1) Barley.csv
2) Plant_growth.csv
3) water_quality.csv

The following external packages are used in this lab. Install them if you have not done so already.

```
install.packages(c("tidyverse", "car"),
  repo = "https://cloud.r-project.org")
```

Finally, try to complete today's lab exercises in pairs and try out pair programming, where one person writes the code and the other person reviews each line as it is written. You can swap roles every 10 minutes or so. Showing someone how you code is an excellent way to learn, and you will be surprised how much you can learn from each other.

Exercise 1: barley (walk-through)

Background

An experiment was designed to compare two varieties of spring barley. Thirty four plots were used, seventeen being randomly allocated to variety A and seventeen to variety B. Unfortunately five plots were destroyed. The yields (t/ha) from the remaining plots were as they appear in the file `Barley.csv`.

Instructions

First, quickly explore the data; then, utilise the **HATPC** process and test the hypothesis that the two varieties give equal yields, assuming that the samples are independent.

HATPC:

- Hypothesis
- Assumptions
- Test (statistic)
- P-value
- Conclusion

> ℹ Level of significance
>
> **The level of significance is usually set at 0.05.** This value is generally accepted in the scientific community and is also linked to Type 2 errors, where choosing a lower significance increases the likelihood of failing to reject the null hypothesis when it is false.

Data exploration

First we load the data and inspect its structure to see if it needs to be cleaned or transformed. The `glimpse()` function is a tidy version of `str()` that provides a quick overview of the data that focuses on the variables, ignoring data attributes.

Try to compare `str()` and `glimpse()` to see what the differences are.

```
barley <- readr::read_csv("data/Barley.csv") # packagename:: before a function lets you access a function without having to load the whole library first
dplyr::glimpse(barley)
```

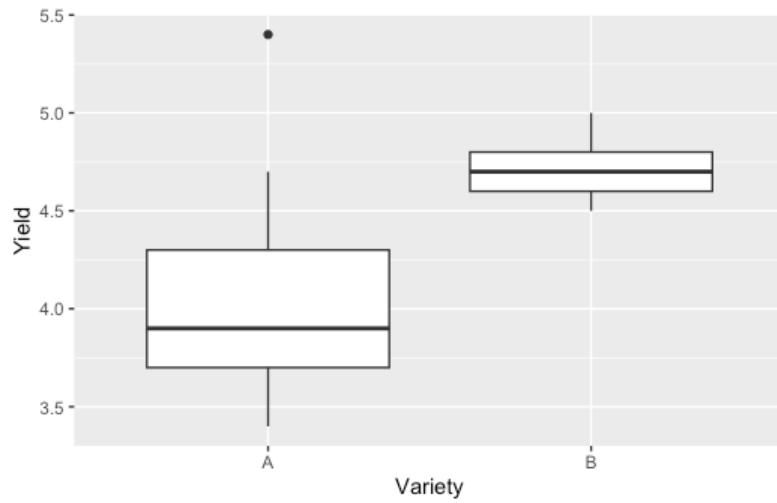
```
Rows: 29
Columns: 2
$ Variety <chr> "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A…
$ Yield   <dbl> 4.6, 4.3, 3.8, 3.4, 3.9, 3.9, 3.9, 4.4, 3.6, 3.6, 4.7, 3.9, 3.…
```

The `Variety` column is a factor with two levels, `A` and `B`, but it is defined as a character. We can convert it to a factor using the `mutate()` function from the `dplyr` package, but it is not necessary for the *t*-test since R will automatically convert it to a factor.

```
library(tidyverse)
barley <- mutate(barley, Variety = as.factor(Variety))
```

Quickly preview the data as a plot to see if there are any trends or unusual observations.

```
barley %>%
  ggplot(aes(x = Variety, y = Yield)) +
  geom_boxplot()
```



A trained eye will anticipate that the data may not meet the assumption of equal variance; however, we will test this assumption later. Otherwise, there appear to be no unusual observations in the data.

Hypothesis

What are the null and alternative hypotheses? We can use the following notation:

$$H_0 : \mu_A = \mu_B$$

$$H_1 : \mu_A \neq \mu_B$$

where μ_A and μ_B are the population means of varieties A and B, respectively.

The equations above are written in LaTeX, a typesetting system that is commonly used in scientific writing. You can learn more about LaTeX here. The raw syntax used to write the equations are shown below:

```
$$H_0: \mu_A = \mu_B$$
$$H_1: \mu_A \neq \mu_B$$
```

It is **important** that when using mathematical symbols to denote the null and alternative hypotheses, you should **always** define what the symbols mean. Otherwise, the reader may not understand what you are referring to.

Why do we always define the null and alternative hypotheses? In complex research projects or when working in a team, it is important to ensure that everyone is on the same page. By defining the hypotheses, you can avoid misunderstandings and ensure that everyone is working towards the same goal as the mathematical notation is clear and unambiguous.

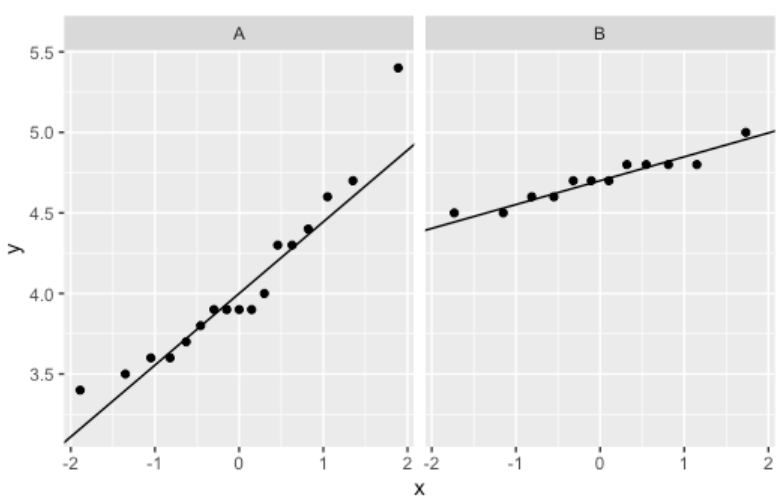
Assumptions

Normality

There are many ways to check for normality. Here we will use the QQ-plot. Use of `ggplot2` is preferred (as a means of practice) but since we are just exploring data, base R functions are not a problem to use.

Using `ggplot2`

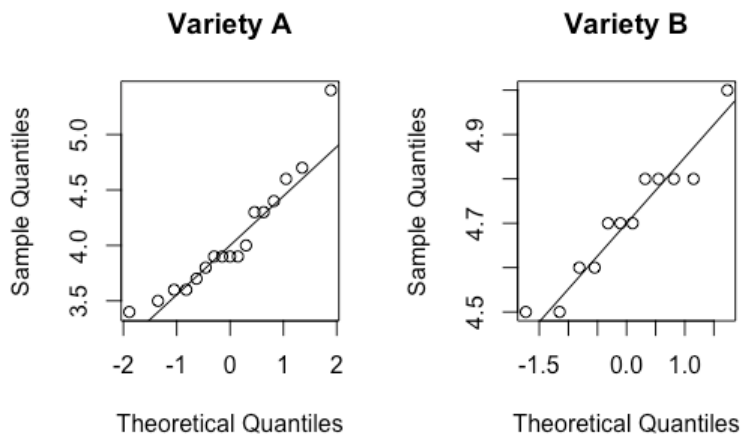
```
ggplot(barley, aes(sample = Yield)) +
  stat_qq() +
  stat_qq_line() +
   facet_wrap(~Variety) #facet_wrap ensures there are separate plots for each variety rather than one plot with all the data in Yield
```



Using base R

```
par(mfrow = c(1, 2))
qqnorm(barley$Yield[barley$Variety == "A"], main = "Variety A") # square brackets to subset the data by variety
```

```r
qqline(barley$Yield[barley$Variety == "A"])
qqnorm(barley$Yield[barley$Variety  ==  "B"],  main  =
"Variety B")
qqline(barley$Yield[barley$Variety == "B"])
```



Question: Do the plots indicate the data are normally distributed?

Answer: Yes, the data appear to be normally distributed as the QQ-plot shows that the data points are close to the line.

Homogeneity of variance

From the boxplot, we can see that there is some indication that the variances are not equal. We can test this assumption using a F-test, Bartlett's test or Levene's test; here we will just use Bartlett's test (look up the different tests for equal variance and pros and cons of each).

H_0 : Variance is equal across all groups
H_1 : Variance is NOT equal across all groups

```r
bartlett.test(Yield ~ Variety, data = barley)
```

```
    Bartlett test of homogeneity of variances

data:  Yield by Variety
Bartlett's  K-squared  =  14.616,  df  =  1,  p-value  =
0.0001318
```

Question: Does the Bartlett's test indicate the two groups have equal variances? What effect will that have on the analysis?

Answer: The two groups have unequal variance (Bartlett's test: $X^2 = 14.6$, $p < 0.01$). This means that we will need to use the Welch's *t*-test, which does not assume equal variances.

Test statistic

We can now calculate the test statistic using the `t.test()` function in R. Since the variances are unequal, we do not have to specify the `var.equal` argument – the default test for `t.test()` is the Welch's *t*-test which does not assume equal variances.

```
t.test(Yield ~ Variety, data = barley)
```

```

    Welch Two Sample t-test

data:  Yield by Variety
t = -4.9994, df = 19.441, p-value = 7.458e-05
alternative hypothesis: true difference in means
between group A and group B is not equal to 0
95 percent confidence interval:
 -0.9293569 -0.3814274
sample estimates:
mean in group A mean in group B
       4.052941        4.708333
```

P-value

Since the p-value is < 0.05, we can reject the null hypothesis that the mean yield of both varieties is equal.

Conclusion

The conclusion needs to be brought into the context of the study. In a scientific report or paper, you would write something like this:

> The mean yield of barley variety A was significantly different from that of variety B ($t = -5.0$, $df = 19.4$, $p < 0.01$).

Exercise 2: plant growth

Background

In a test of a particular treatment aimed at inducing growth, 20 plants were grouped into ten pairs so that the two members of each

pair were as similar as possible. One plant of each pair was chosen randomly and treated; the other was left as a control. The increases in height (in centimetres) of plants over a two-week period are given in the file Two week plant heights. We wish to compare whether the treatment is actually inducing improved growth, as compared to the control.

Instructions

Here, we have two samples, and the samples are paired as it is a before-after experiment (the data are NOT independent). So we'd like to conduct a paired *t*-test.

For paired *t*-tests the analysis is performed as a 1-sample *t*-test on the difference between each pair so the only assumption is the normality assumption.

Copy the structure below and perform your analysis in your document.

```
## Exercise 2: plant growth
### Data exploration
### Hypothesis
### Assumptions
#### Normality
#### Homogeneity of variance (if required)
### Test statistic
### P-value
### Conclusion
```

Note 1. The data is *not* tidy. The code below will convert the data to the long format and assign it to `tidy_plant` which can be useful for ggplot. Note 2. The data are paired so we look at the difference when testing hypotheses. Note 3. We cannot perform a paired t.test on data in long format - this is to ensure the data remains "tied/paired" i.e. Floris before heart rate doesn't get mixed up with Liana's after heart rate.

```
plant_growth        <-        readr::read_csv("data/
Plant_growth.csv")

tidy_plant <- plant_growth %>%
  pivot_longer(cols = c(Treated, Control), names_to =
"Treatment", values_to = "Height")
```

Answer 2

Data exploration

```
#Boxplot
ggplot(tidy_plant, aes(x = Treatment, y = Height)) +
  geom_boxplot()
```

The boxplot shows no unusual observations.

Hypothesis

$$H_0 : \mu_{\text{treated}} = \mu_{\text{control}}$$

$$H_1 : \mu_{\text{treated}} \neq \mu_{\text{control}}$$

where μ_{treated} and μ_{control} are the mean increses in height of the treated and control plants, respectively.

Assumptions

Normality

```
#qqplots
plant_growth$Diff    <-    plant_growth$Treated    -
plant_growth$Control
ggplot(plant_growth, aes(sample = Diff)) +
  stat_qq() +
  stat_qq_line()
```

The QQ-plots show that the data are normally distributed, but there are some outliers worth checking and the dataset is small. A formal test for normality will give us a better idea of whether the assumption is met.

```
# perform test on groups
shapiro.test(plant_growth$Diff)
```

The Shapiro-Wilk test indicates that the data are normally distributed.

Homogeneity of variance

There is **no need** to test for homogeneity of variance in a paired *t*-test!

Test statistic

Remember you must indicate paired = TRUE!!

```
t.test(plant_growth$Treated,
  plant_growth$Control,
  paired = TRUE
)
```

P-value

The p-value is 0.03 which is less than 0.05, so we reject the null hypothesis that the mean increase in height of the treated and control plants is equal.

Conclusion

The mean increase in height of the treated plants is significantly different from that of the control plants ($t = 2.5, df = 9, p = 0.03$).

Exercise 3: turbidity

Background

A new filtering process was installed at a dam which provided drinking water for a nearby town. To check on its success, a number of water samples were taken at random times and locations in the weeks before and after the process was installed. The following are the turbidity values (units = NTU) of the water samples.

Instructions

Now we consider further examples of a two-sample *t*-test, but where the assumption of equal variance and normality may not be met for the raw data. Sometimes after applying a data transformation the analysis can proceed assuming equal variances – but always check after a data transformation.

The data can be read with the code below:

```
water_quality <- read_csv("data/water_quality.csv")
```

For data transformation, you may need to create a new variable in your dataset to store the transformed data. For example, to create a new variable `TurbLog10` that stores the $\log_{10}$ transformed turbidity values, you can use the following code:

```
water_quality$TurbLog10
<- log10(water_quality$Turbidity)
```

To interpret the results for your conclusions, you may need to back-transform the mean and/or confidence interval values. To back transform $\log_{10}$ data you use:

$$10^{\text{mean or CI}}$$

To back-transform natural log, $\log_e$, you use:

$$e^{\text{mean or CI}}$$

Answer 3

Data exploration

```
#Read in data
water_quality <- read_csv("data/water_quality.csv")

#Make boxplot
ggplot(water_quality, aes(x = Time, y = Turbidity)) +
  geom_boxplot()
```

The boxplot shows skewing of the data, and outliers, a strong indication that the data will have issues meeting the assumption of normality. It is also likely that the variances are not equal as the boxplot shows different spreads.

Hypothesis

$$H_0 : \mu_{\text{before}} = \mu_{\text{after}}$$

$$H_1 : \mu_{\text{before}} \neq \mu_{\text{after}}$$

where μ_{before} and μ_{after} are the mean turbidity values before and after the new filtering process was installed.

Assumptions

Normality: part 1

```
#qqplots
ggplot(water_quality, aes(sample = Turbidity)) +
  stat_qq() +
  stat_qq_line() +
  facet_wrap(~Time)
```

The QQ-plots show that the data are not normally distributed. Transformation of the data is required – there is no need to look at the equal variances just yet.

Transform data

We will use the natural log transformation to normalise the data.

```
water_quality$TurbLog10
<- log10(water_quality$Turbidity)
```

Normality: part 2

```
ggplot(water_quality, aes(sample = TurbLog10)) +
  stat_qq() +
  stat_qq_line() +
  facet_wrap(~Time)
```

The QQ-plots show that the data look better. We can now perform a Shapiro-Wilk test to confirm this.

```
tapply(water_quality$TurbLog10,   water_quality$Time,
shapiro.test)
```

The Shapiro-Wilk test indicates that the data are normally distributed.

Homogeneity of variance

We check the assumption of equal variance after the transformation.

H_0 : Variance is equal across all groups
H_1 : Variance is NOT equal across all groups

```
bartlett.test(TurbLog10 ~ Time, data = water_quality)
```

The Bartlett's test indicates that the variances are equal. We can now perform the *t*-test on the transformed data.

Test statistic

```
t.test(TurbLog10  ~  Time,  data  =  water_quality,
var.equal = TRUE)
```

P-value

The p-value is 0.007, which is less than 0.05, so we reject the null hypothesis that the mean turbidity values before and after the new filtering process was installed are equal.

Conclusion

We want to make a biological conclusion based on the results. To do this, we need to back-transform the mean difference and the confidence intervals so that the units are in NTU, not $\log_{10}$ NTU.

Mean difference

```
10^(1.514274 - 0.683656)
```

Lower and upper CI

```
10^(0.359140)
10^(1.302095)
```

The mean turbidity values before and after the new filtering process was installed are significantly different ($t = -3.7$, $df = 17$, $p = 0.007$). The new filtering process has effectively reduced turbidity by 7 times on average. We are 95% confident that the new filtering process will reduce water turbidity by between 2.3 and 20 NTU.

Wait! Why 7 times? Recall the log law: log10(A) - log10(B) = log10(A/B)

So the difference in the logged means: 1.514274 - 0.683656 = log10(A/B)

And to backtransform, we'd use 10^(1.514274 - 0.683656) = A/B

Therefore the backtransformation of the difference in the logged means returns **the ratio of the first mean to the second mean**. This will be covered in next week's lecture.

Bonus take home exercises

Use the HATPC framework to answer the following questions

1. Hypothesis
2. Assumptions
3. Test (statistic)
4. P-value
5. Conclusion

Exercise 1: Tooth growth

Tooth GRowth is an inbuilt dataset that shows the effect of vitamin c in Guinea pig tooth growth. It has three variables:

- len = tooth length
- supp = type of supplement (Orange juice or ascorbic acid)

- dose = mg/day given to the guinea pigs

```
head(ToothGrowth)
str(ToothGrowth)
```

Using the HATPC framework, test whether the type of supplent affects tooth length.

Solution

1. Hypothesis

   H_0: $\mu_{OJ} = \mu_{VC}$

   H_1: $\mu_{OJ} \neq \mu_{VC}$

2. Assumptions

Normality

```
library(tidyverse)
# qqplot
ggplot(ToothGrowth, aes(sample = len)) +
  stat_qq() +
  stat_qq_line() +
  facet_wrap(~supp)
```

The data appear to be normally distribute, as most of the dots follow the line

Homogeneity of variance

```
#Boxplots
ggplot(ToothGrowth, aes(x = supp, y = len))+
  geom_boxplot()
```

```
# Barttlet test
bartlett.test(len~supp, data = ToothGrowth)
```

The two groups have equal variance

3. Test statistic

```
t.test(len~supp, var.equal = TRUE, data = ToothGrowth)
```

4. P-value

   Since the p-value > 0.05, we retain the null hypothesis

5. Conclusion

   The mean tooth length was the same for each treatment group. The type of supplement does not have a significant effect on guinea pig tooth length

Exercise 2: Adelie penguin bill length

For this exercise, we will be using a subset of the palmer penguins dataset.

```
library(palmerpenguins)
library(tidyverse)

adelie <-penguins%>%
  filter(species == "Adelie")%>%
  na.omit()%>%
  droplevels()

head(adelie)
```

Using the HATPC framework, test whether male and female penguins have the same length bill

Solution

1. Hypothesis

   H_0: $\mu_{fbilllength} = \mu_{mbilllength}$

   H_1: $\mu_{fbilllength} \neq \mu_{mbilllength}$

2. Assumptions

Normality

```
library(tidyverse)
```

```r
# qqplot
ggplot(adelie, aes(sample = bill_length_mm))+
  geom_qq()+
  geom_qq_line()+
  facet_wrap(~sex)
```

```r
#shapiro.test
tapply(adelie$bill_length_mm,
adelie$sex, shapiro.test)
```

The two groups appear to be normally distributed, both according to the qqplot and the shapiro test

Homogeneity of variance

```r
#boxplot
ggplot(adelie, aes(x = sex, y = bill_length_mm))+
  geom_boxplot()
```

```r
#barttlet test
bartlett.test(bill_length_mm~sex, data = adelie)
```

The two groups have equal variance

3. Test (statistic)

```r
# t.test
t.test(bill_length_mm~sex,  var.equal  =  TRUE,data  =
adelie)
```

4. P-value

Since p< 0.05, we reject the null hypothesis

5. Conclusion

Male and female adelie penguins have different mean bill legths. Males have longer beaks than females

Exercise 3: Penguin body mass

For this exercise, we will use a subset of the palmer penguins data set again. This time, we will be comparing two different penguin species.

```
library(palmerpenguins)

penguins2<- penguins%>%
  filter(species != "Adelie")%>%
  na.omit()%>%
  droplevels()
```

Using the HATPC framework, test whether chinstrap and gentoo penguins have different body masses.

Solution

1. Hypothesis

H_0: $\mu_{chinstrap} = \mu_{gentoo}$

H_1: $\mu_{chinstrap} \neq \mu_{gentoo}$

2. Assumptions

Normality

```
library(tidyverse)

#qqplot

ggplot(penguins2, aes(sample = body_mass_g))+
  geom_qq()+
  geom_qq_line()+
  facet_wrap(~species)
```

```
#shapiro test

tapply(penguins2$body_mass_g,      penguins2$species,
shapiro.test)
```

The data appear to be normally distributed. The point in the qq plot follow the line pretty closely, and the shapiro tests also indicate a normal distribution

Homogeneity of variance

```
# barttlettest
bartlett.test(penguins2$body_mass_g~penguins2$species)
```

According to the BArttlet test, the two samples do not have equal variances

3. Test (statistic)

```
t.test(body_mass_g~species, var.equal = FALSE, data =
penguins2)
```

4. P-value

Since p>0.05, we reject the null hypothesis

5. Conclusion

Chinstrap and gentoo penguins have different mean body masses. Gentoo pneguins are heavier than chinstrap penguins on average.